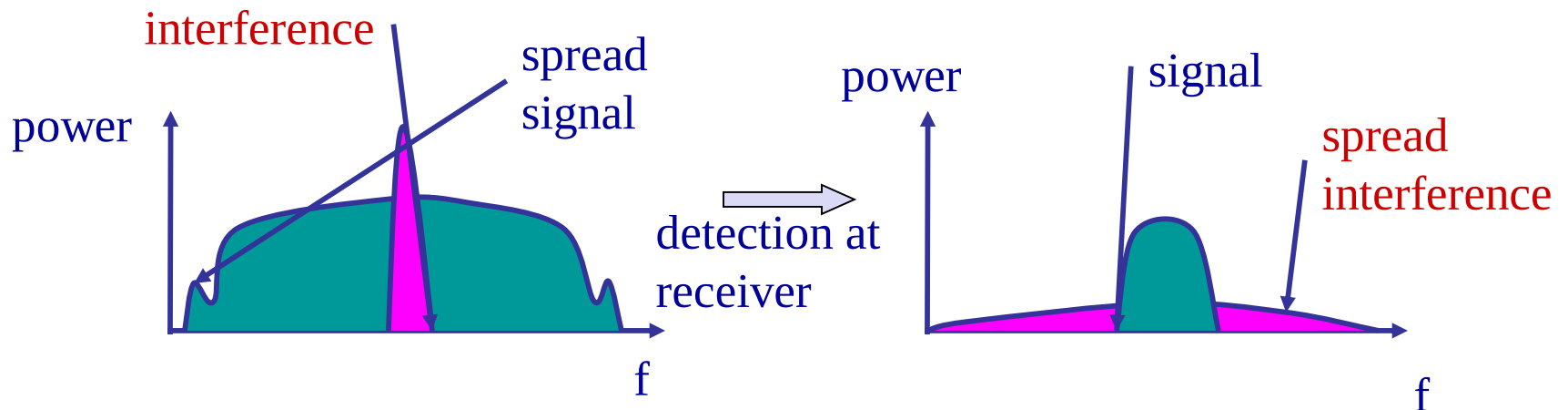

Click to add Text

SPREAD SPECTRUM (**DSSS, FHSSS**)

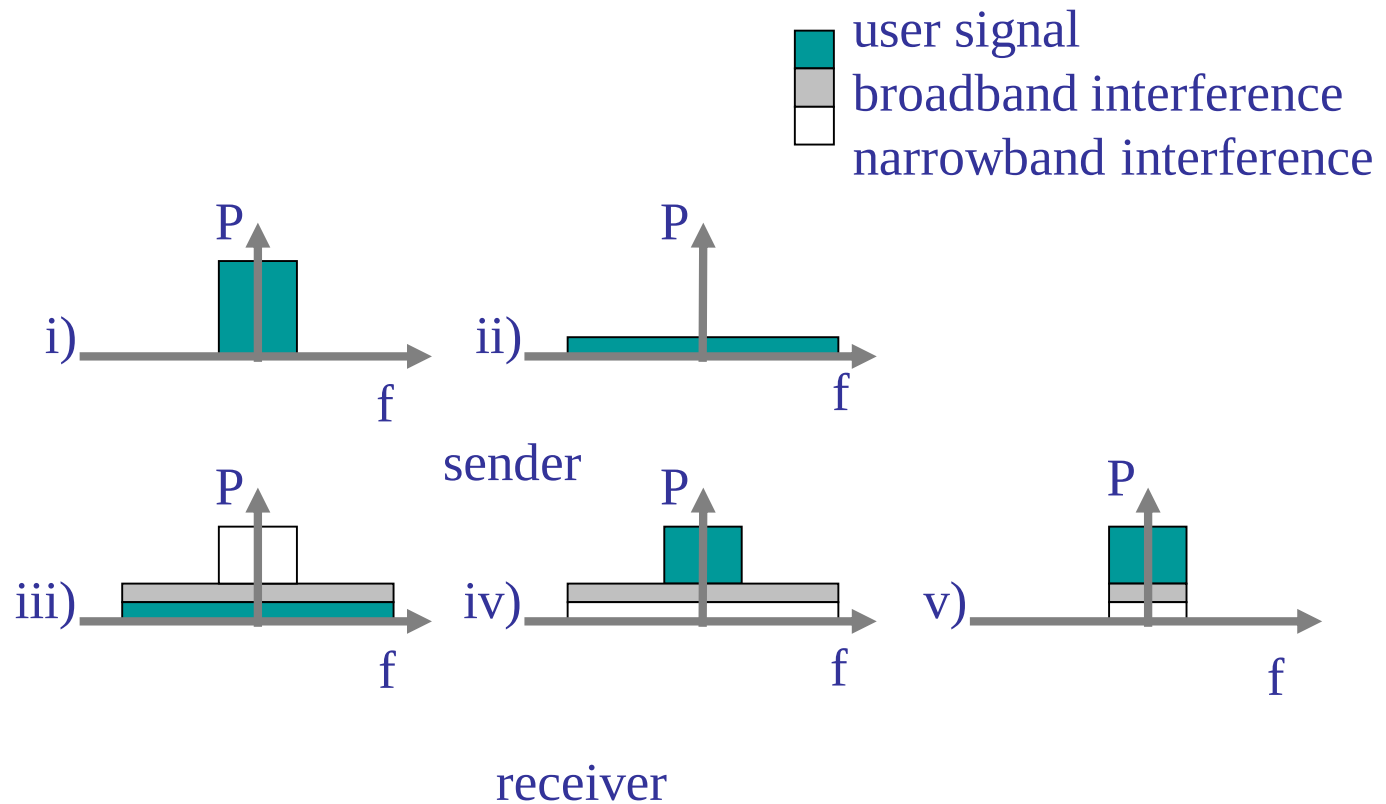
Spread Spectrum Technology

Problem of radio transmission: frequency dependent fading can wipe out narrow band signals for duration of the interference

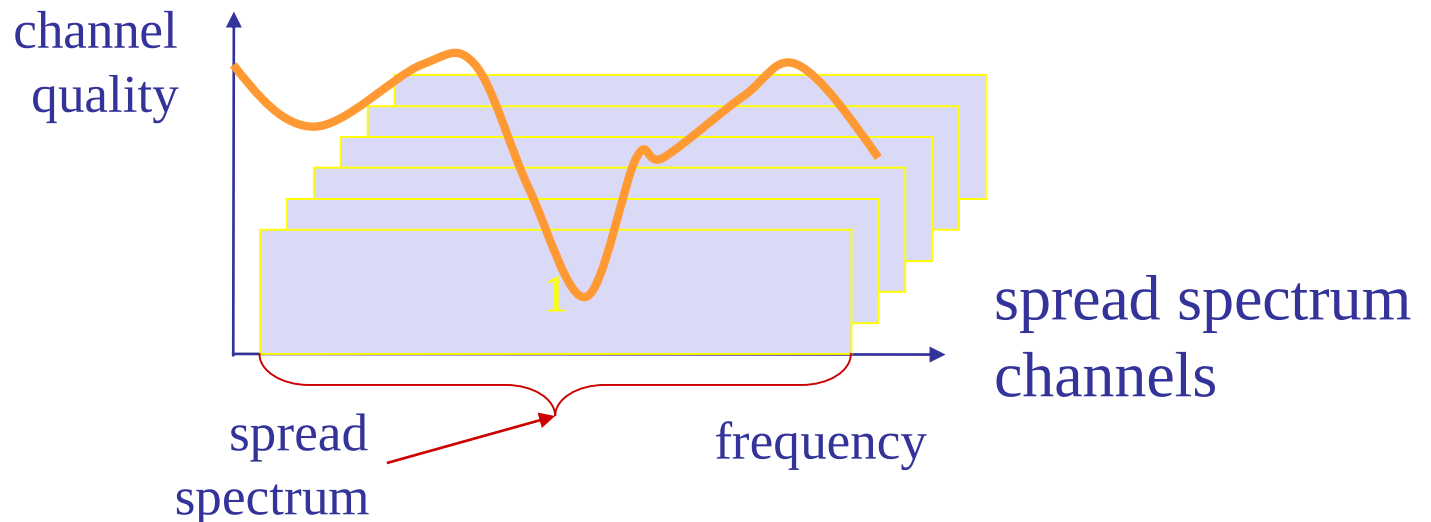
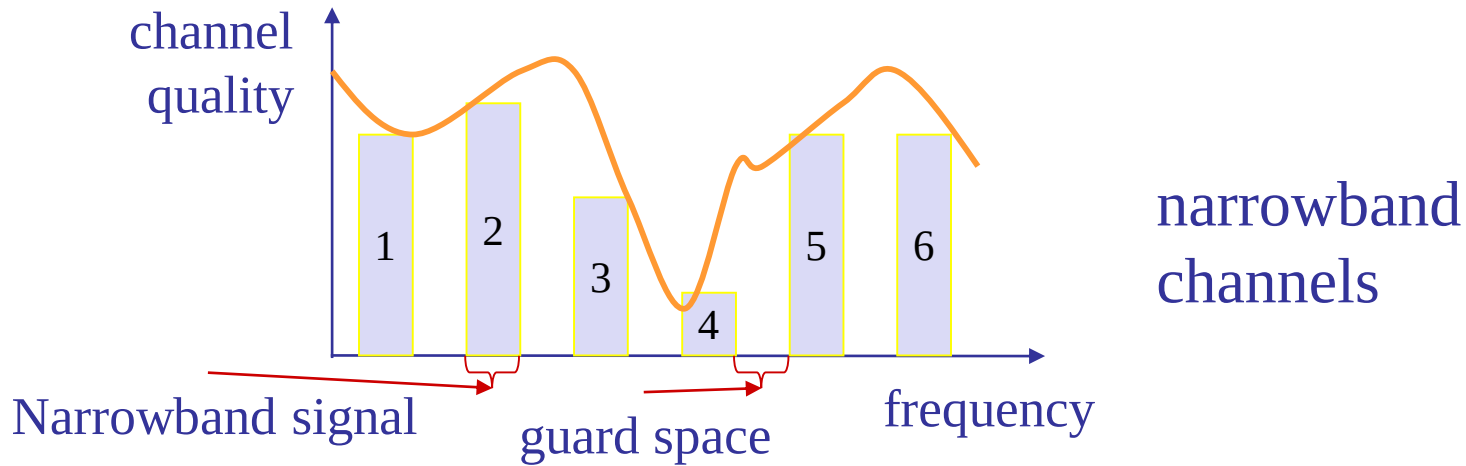
Solution: spread the narrow band signal into a broad band signal using a special code



Effects of spreading and interference



Spreading and frequency selective fading



What is Spread Spectrum?

Spread spectrum techniques are methods by which electromagnetic energy generated in a particular bandwidth is deliberately spread in the frequency domain, resulting in a signal with a wider bandwidth

Spread spectrum methods:

- Frequency hopping spread spectrum
- Direct sequence spread spectrum

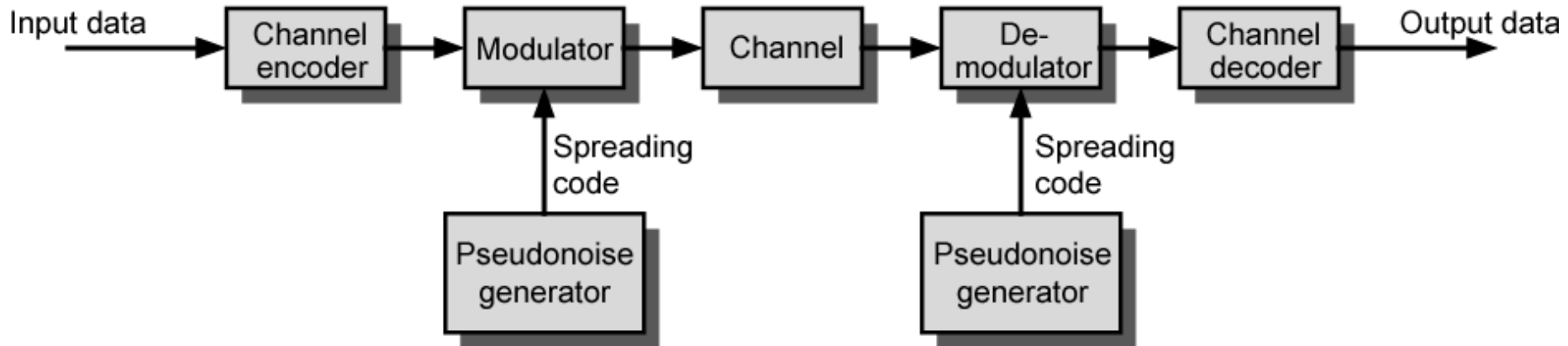
Spread Spectrum

- Analog or digital data
- Analog signal
- Spread data over wide bandwidth
- Makes jamming and interception harder
- Frequency hopping
 - Signal broadcast over seemingly random series of frequencies
- Direct Sequence
 - Each bit is represented by multiple bits in transmitted signal
 - Chipping code

Spread Spectrum Concept

- Input fed into channel encoder
 - Produces narrow bandwidth analog signal around central frequency
- Signal modulated using sequence of digits
 - Spreading code/sequence
 - Typically generated by pseudonoise/pseudorandom number generator
- Increases bandwidth significantly
 - Spreads spectrum
- Receiver uses same sequence to demodulate signal
- Demodulated signal fed into channel decoder

General Model of Spread Spectrum System



Gains

- Immunity from various noise and multipath distortion
 - Including jamming
- Can hide/encrypt signals
 - Only receiver who knows spreading code can retrieve signal
- Several users can share same higher bandwidth with little interference
 - Cellular telephones
 - Code division multiplexing (CDM)
 - Code division multiple access (CDMA)

Pseudorandom Numbers

- Generated by algorithm using initial seed
- Deterministic algorithm
 - Not actually random
 - If algorithm good, results pass reasonable tests of randomness
- Need to know algorithm and seed to predict sequence

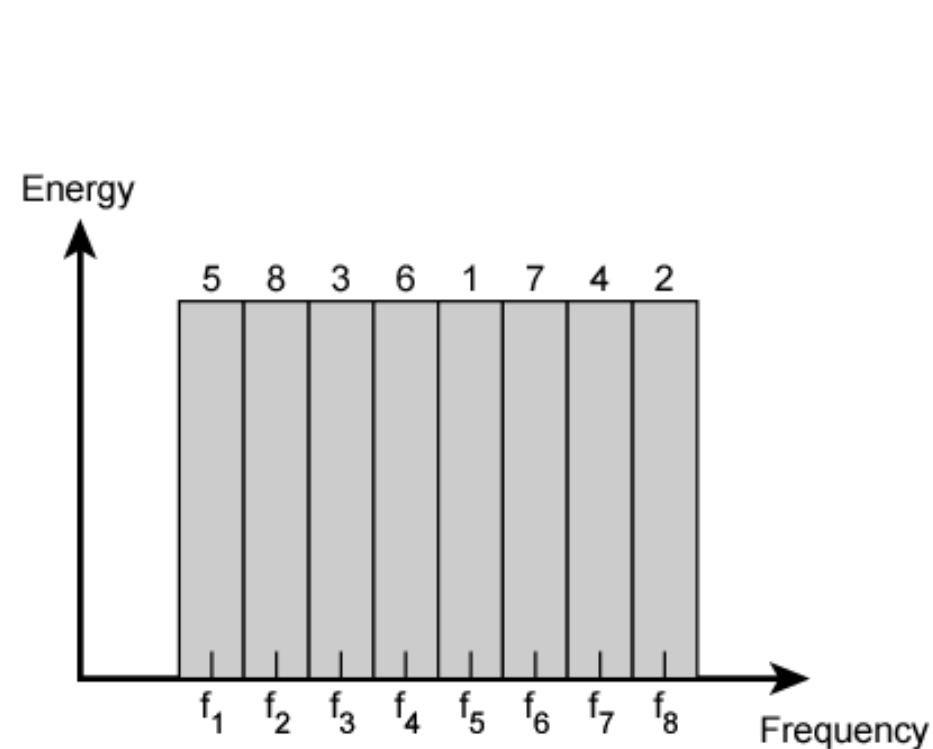
Frequency Hopping Spread Spectrum (FHSS)

- Signal broadcast over seemingly random series of frequencies
- Receiver hops between frequencies in sync with transmitter
- Eavesdroppers hear unintelligible blips
- Jamming on one frequency affects only a few bits

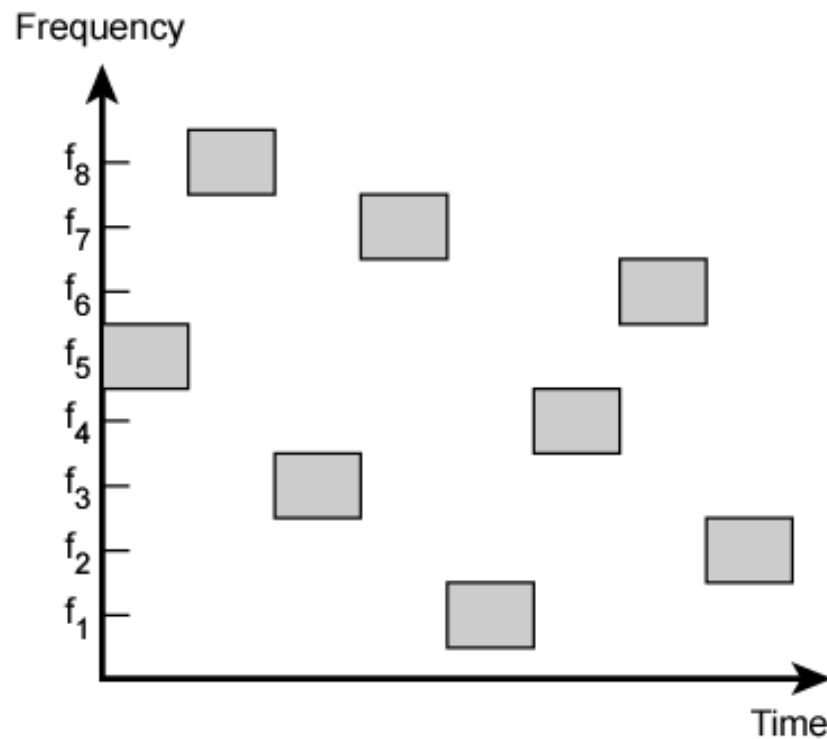
Basic Operation

- Typically 2^k carriers frequencies forming 2^k channels
- Channel spacing corresponds with bandwidth of input
- Each channel used for fixed interval
 - 300 ms in IEEE 802.11
 - Some number of bits transmitted using some encoding scheme
 - May be fractions of bit (see later)
 - Sequence dictated by spreading code

Frequency Hopping Example

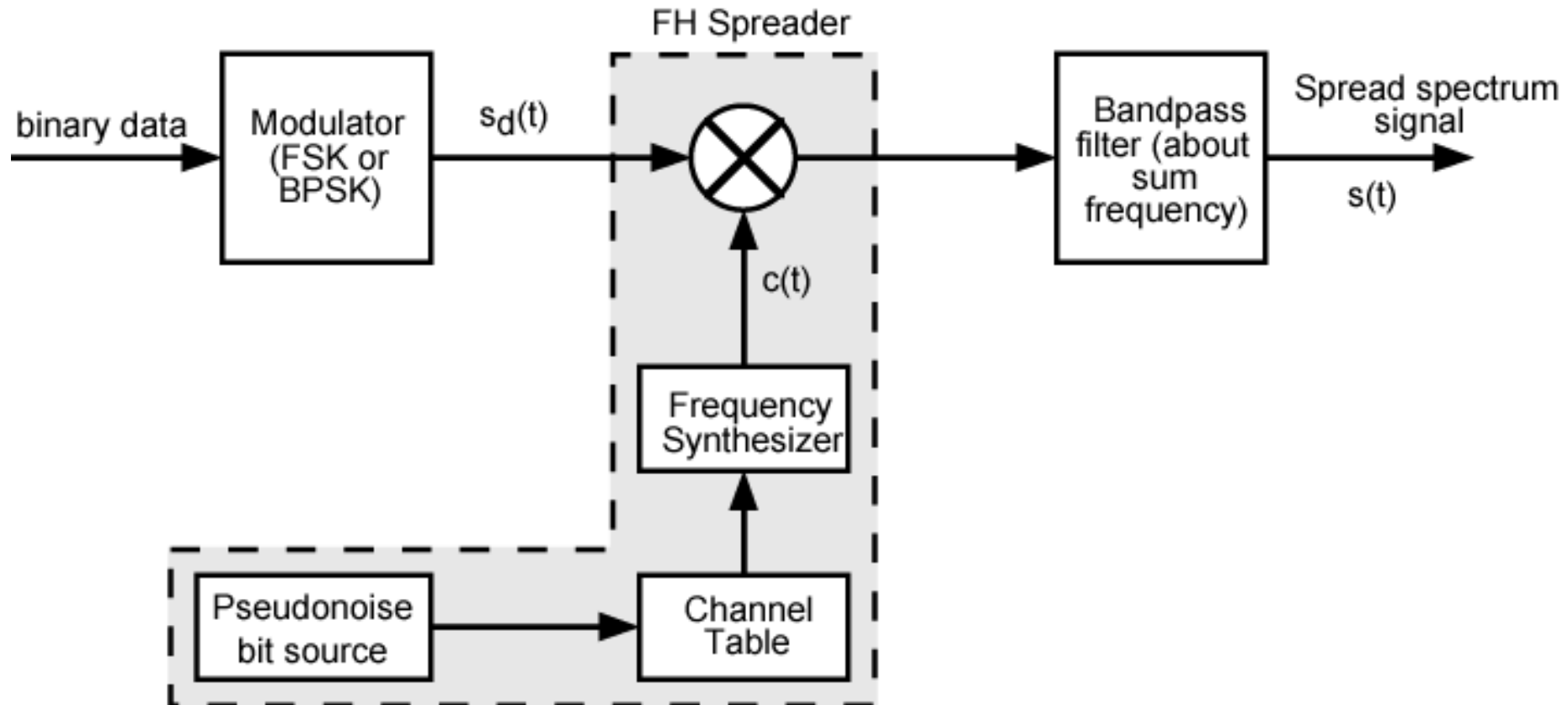


(a) Channel assignment

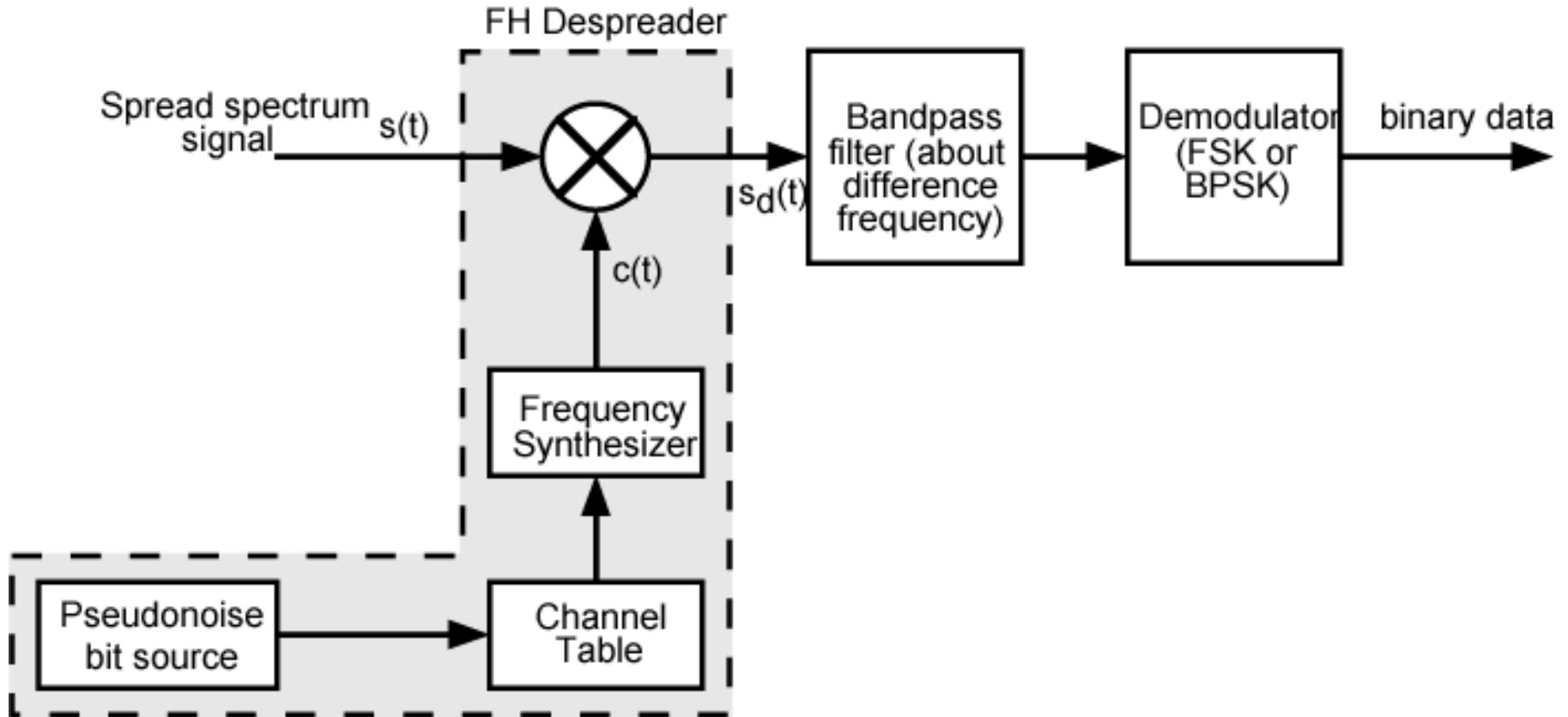


(b) Channel use

Frequency Hopping Spread Spectrum System (Transmitter)



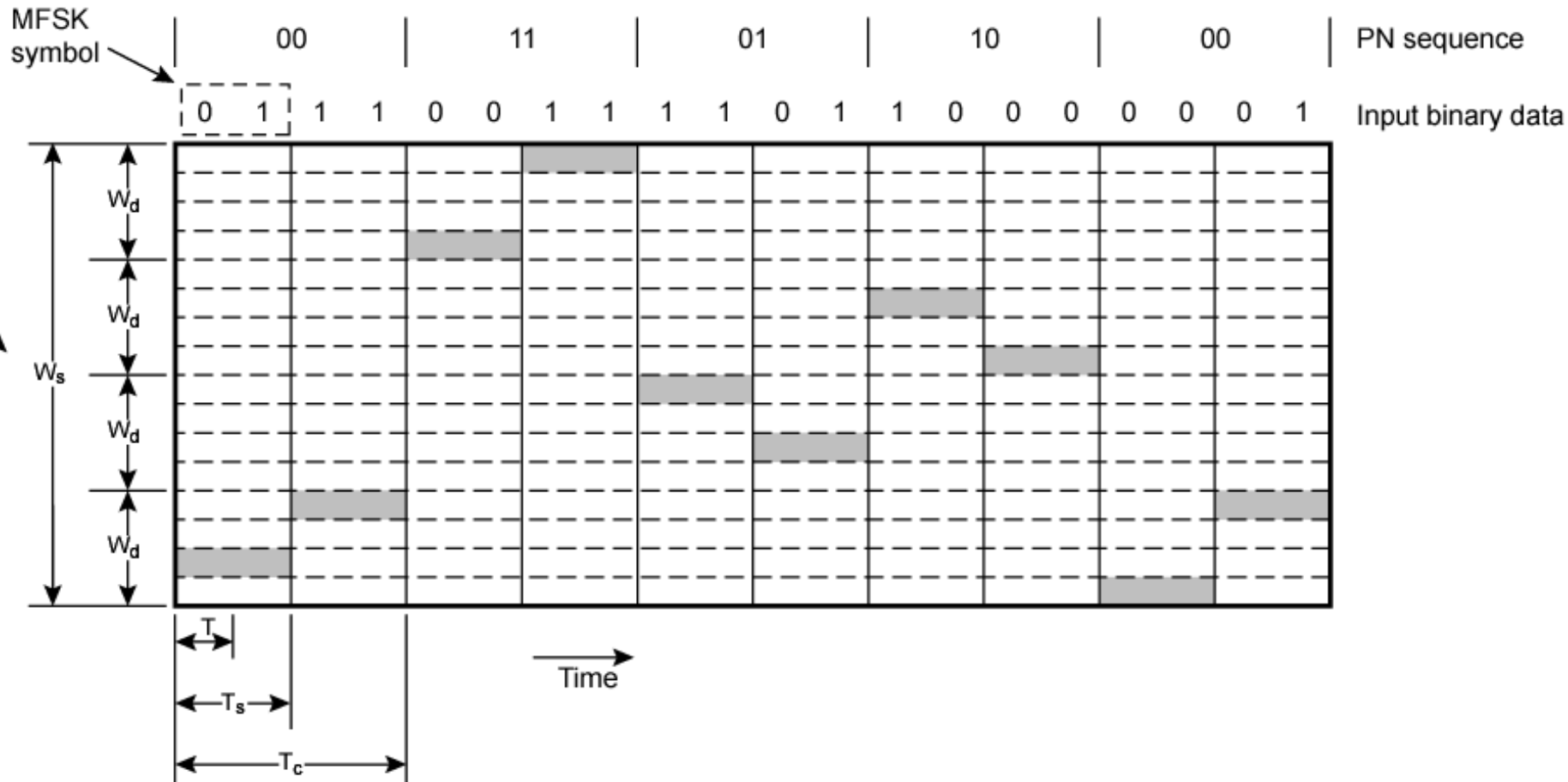
Frequency Hopping Spread Spectrum System (Receiver)



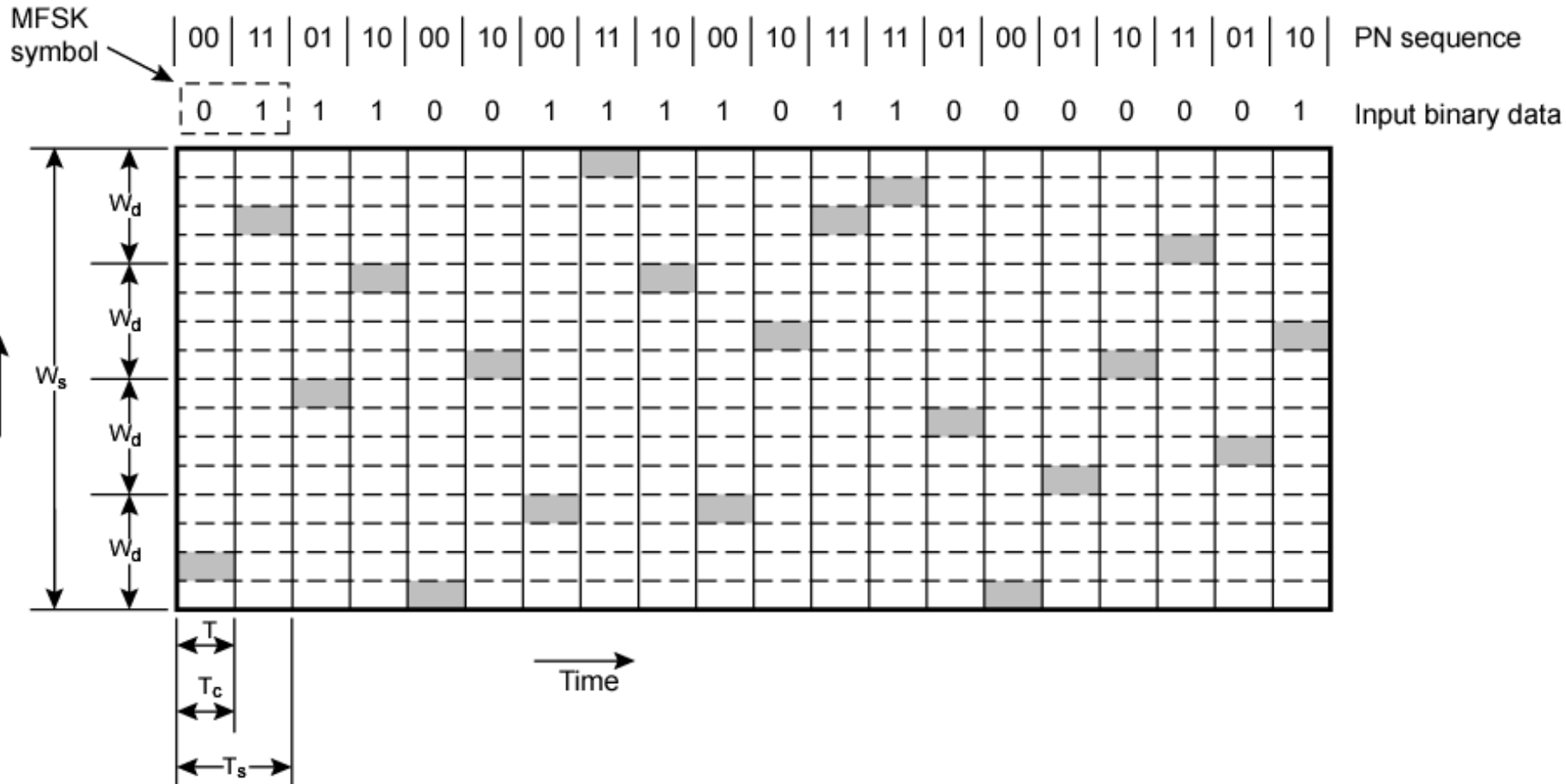
Slow and Fast FHSS

- Frequency shifted every T_c seconds
- Duration of signal element is T_s seconds
- Slow FHSS has $T_c \geq T_s$
- Fast FHSS has $T_c < T_s$
- Generally fast FHSS gives improved performance in noise (or jamming)

Slow Frequency Hop Spread Spectrum Using MFSK (M=4, k=2)



Fast Frequency Hop Spread Spectrum Using MFSK (M=4, k=2)



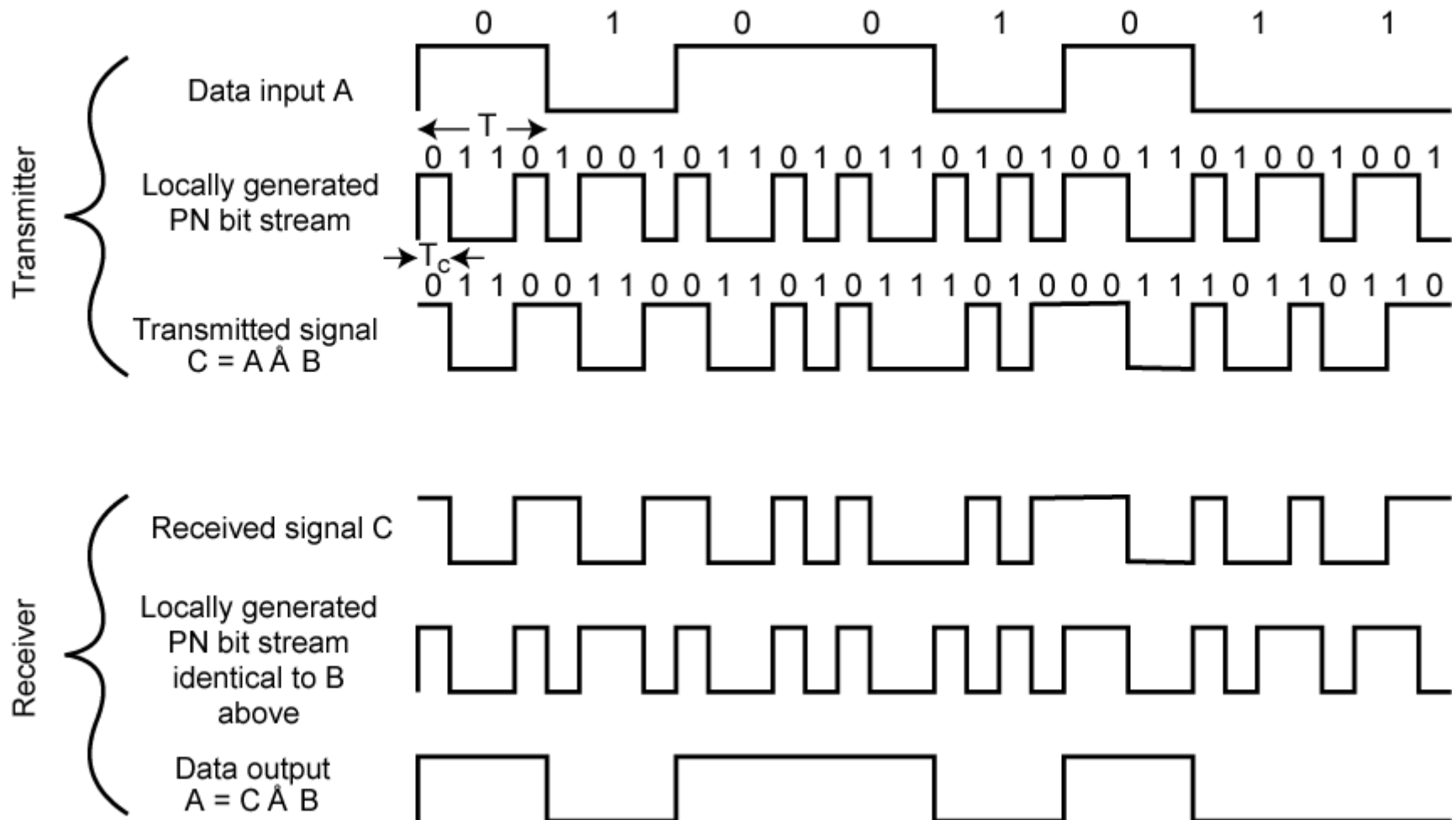
FHSS Performance Considerations

- Typically large number of frequencies used
 - Improved resistance to jamming

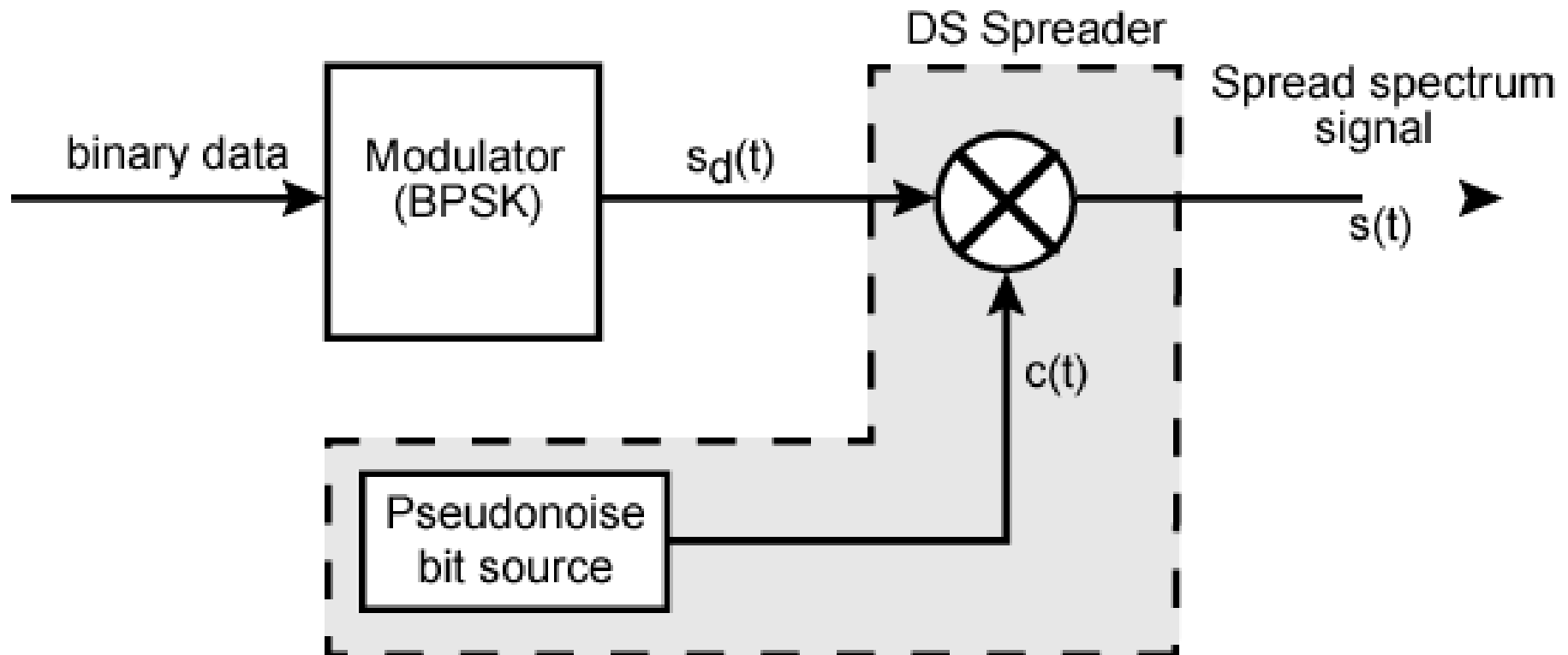
Direct Sequence Spread Spectrum (DSSS)

- Each bit represented by multiple bits using spreading code
- Spreading code spreads signal across wider frequency band
 - In proportion to number of bits used
 - 10 bit spreading code spreads signal across 10 times bandwidth of 1 bit code
- One method:
 - Combine input with spreading code using XOR
 - Input bit 1 inverts spreading code bit
 - Input zero bit doesn't alter spreading code bit
 - Data rate equal to original spreading code
- Performance similar to FHSS

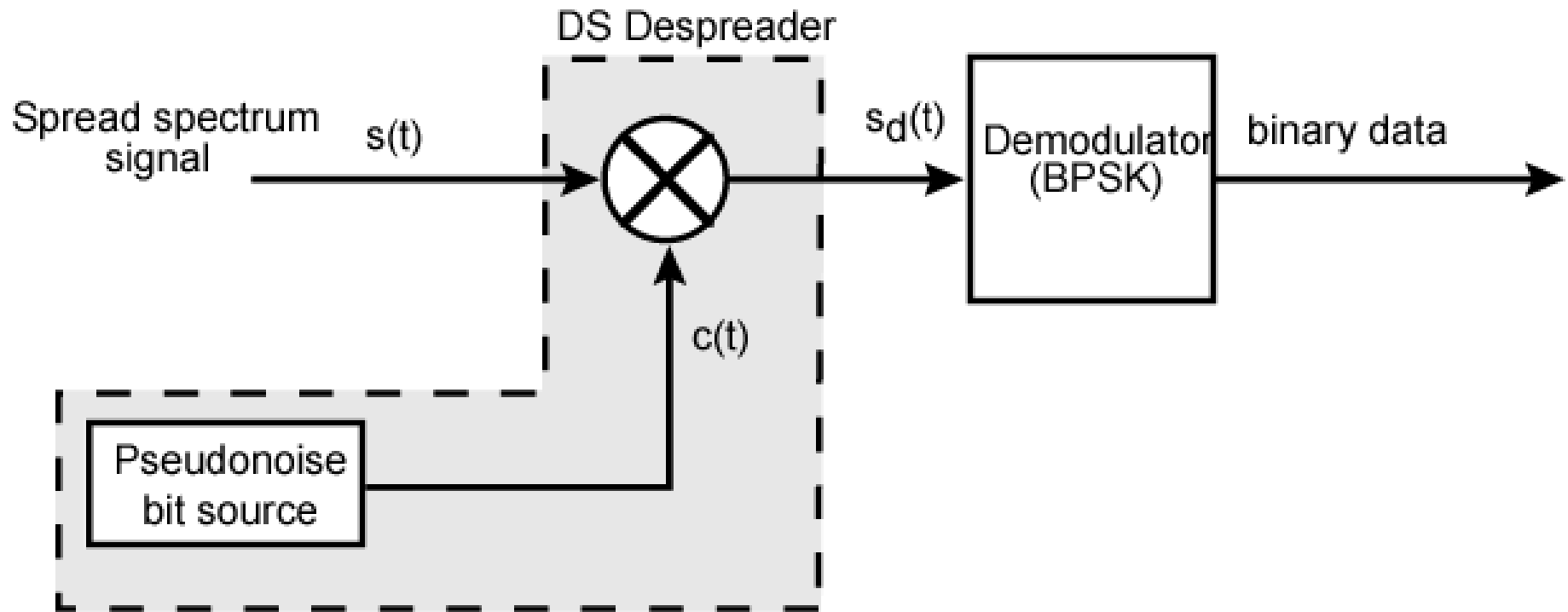
Direct Sequence Spread Spectrum Example



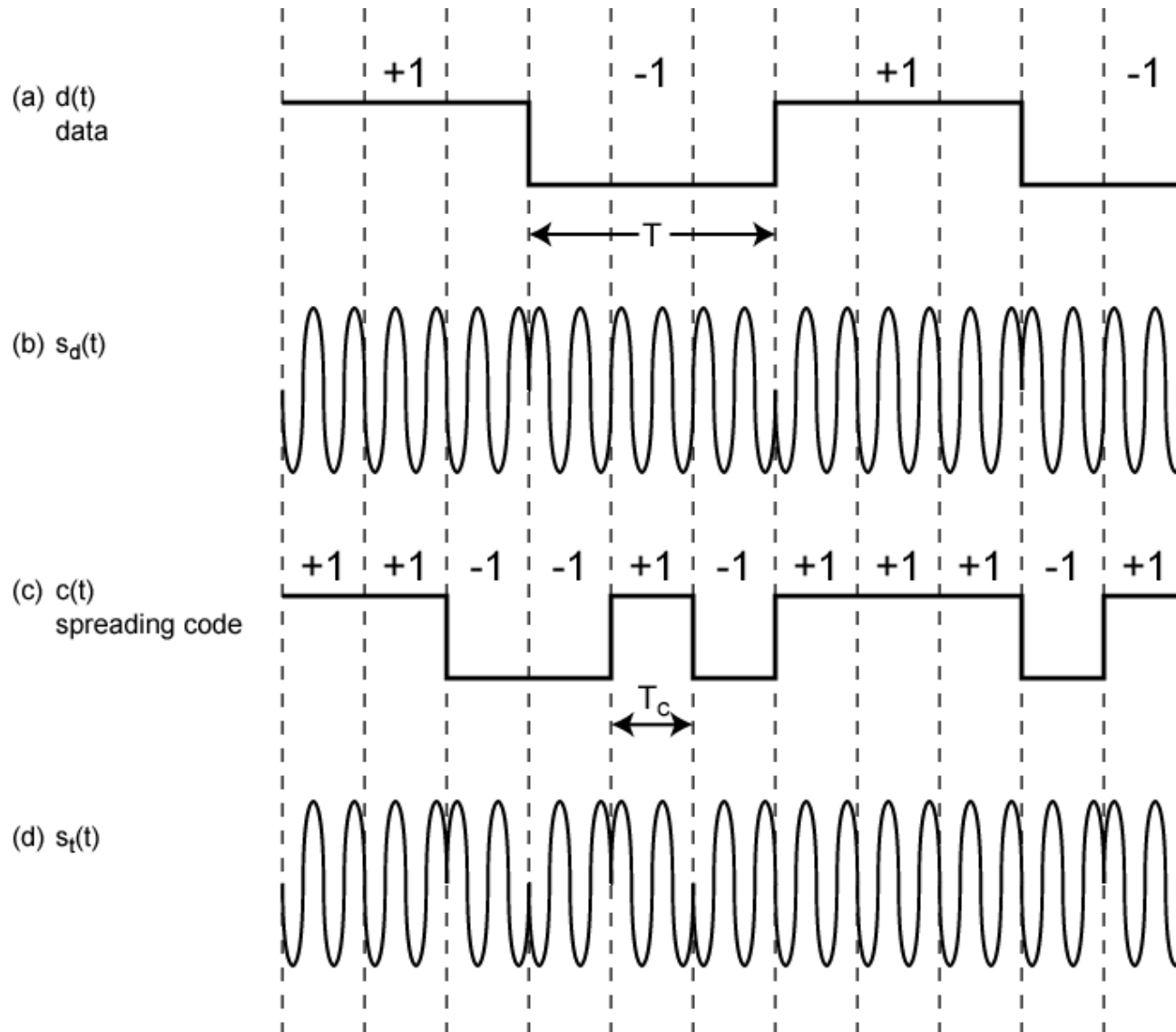
Direct Sequence Spread Spectrum Transmitter



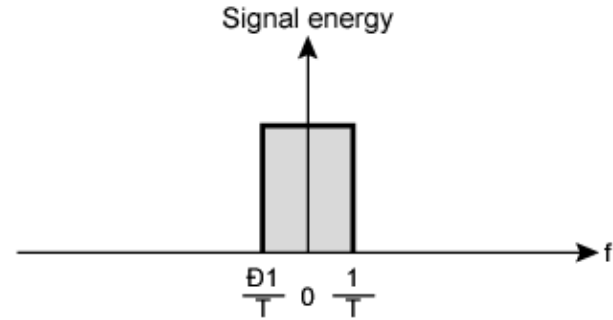
Direct Sequence Spread Spectrum Transmitter



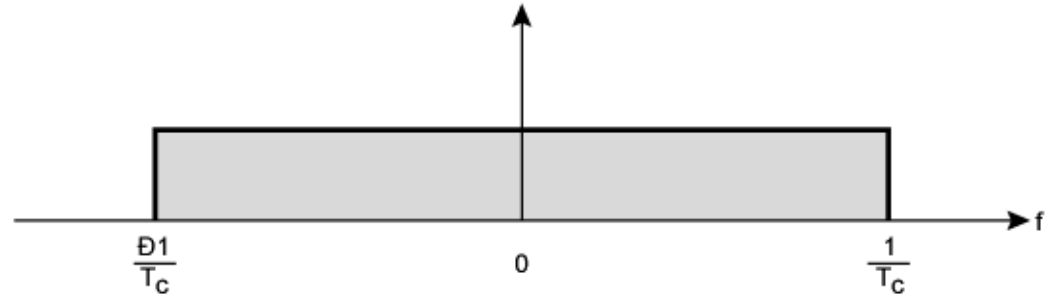
Direct Sequence Spread Spectrum Using BPSK Example



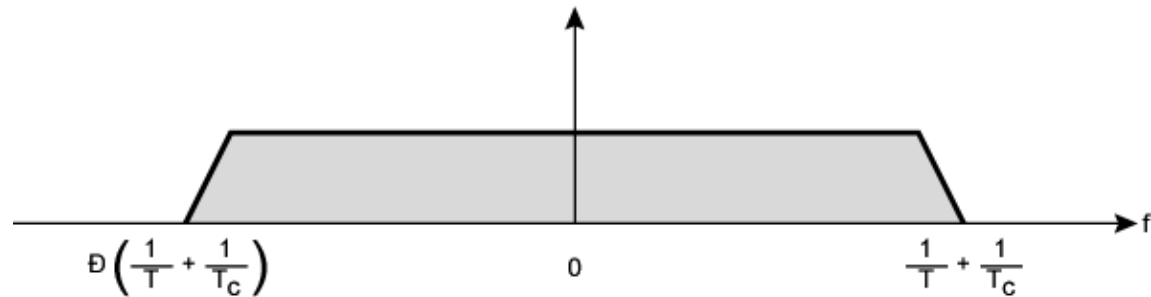
Approximate Spectrum of DSSS Signal



(a) Spectrum of data signal



(b) Spectrum of pseudonoise signal



(c) Spectrum of combined signal

CDMA in DSSS Environment

- n users each using different orthogonal PN sequence
- Modulate each users data stream
 - Using BPSK
- Multiply by spreading code of user

CDMA in a DSSS Environment

